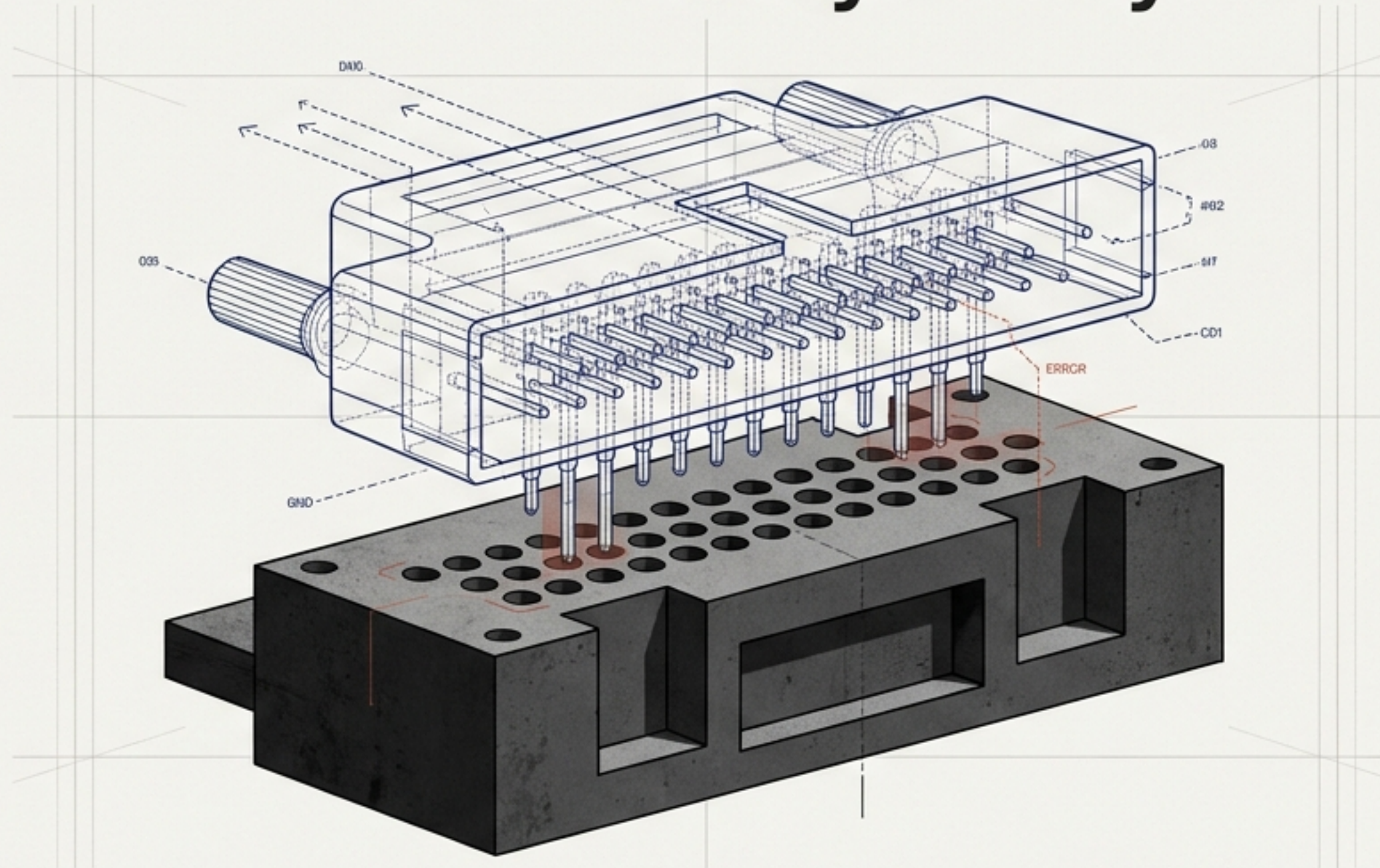


Bridging the gap between the imagined API and binary reality.



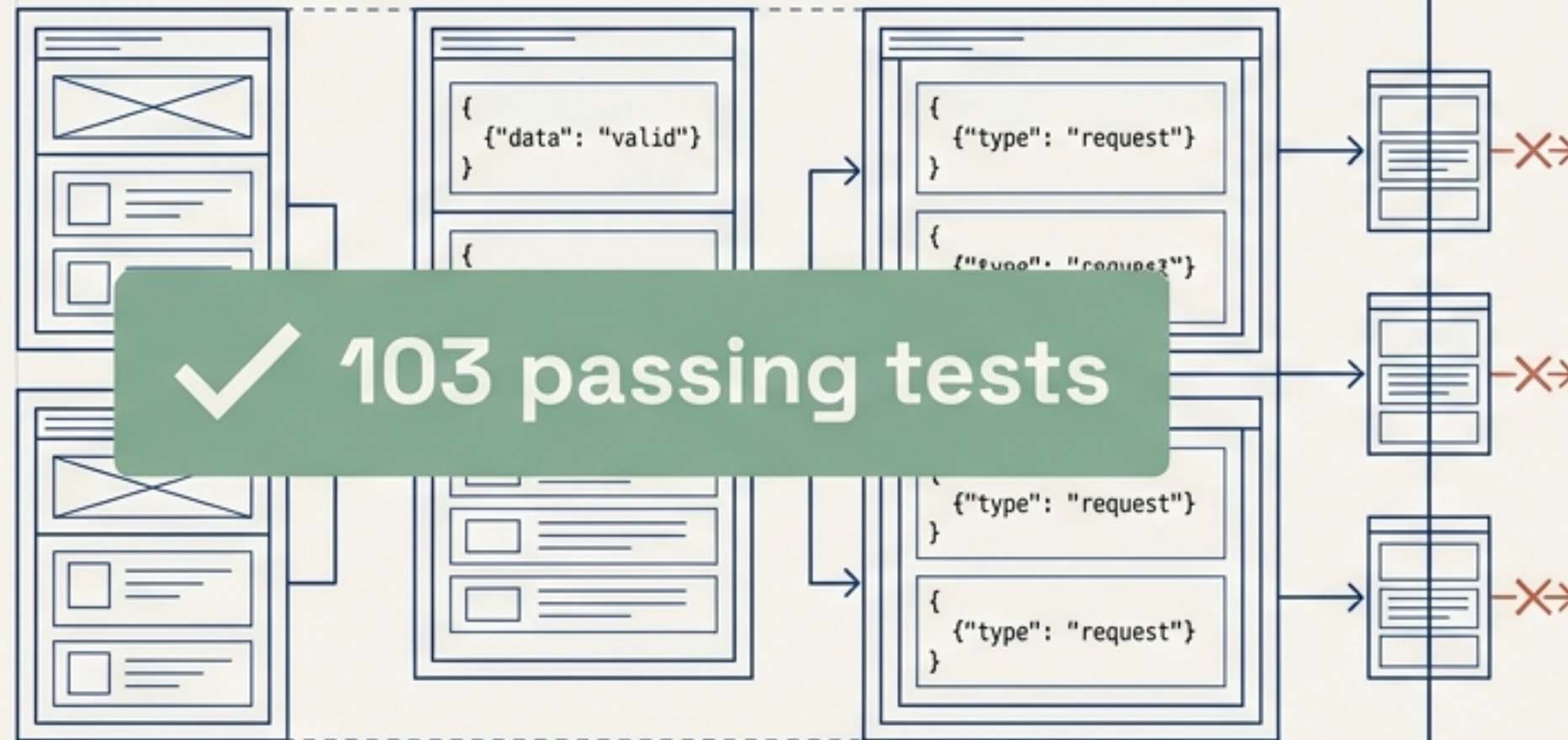
OrchestKit: The Convergence Audit.

Conducted 2026-08-08 across 42 agents and 5 parallel audits.

34 Bugs = 1 Defect x 34

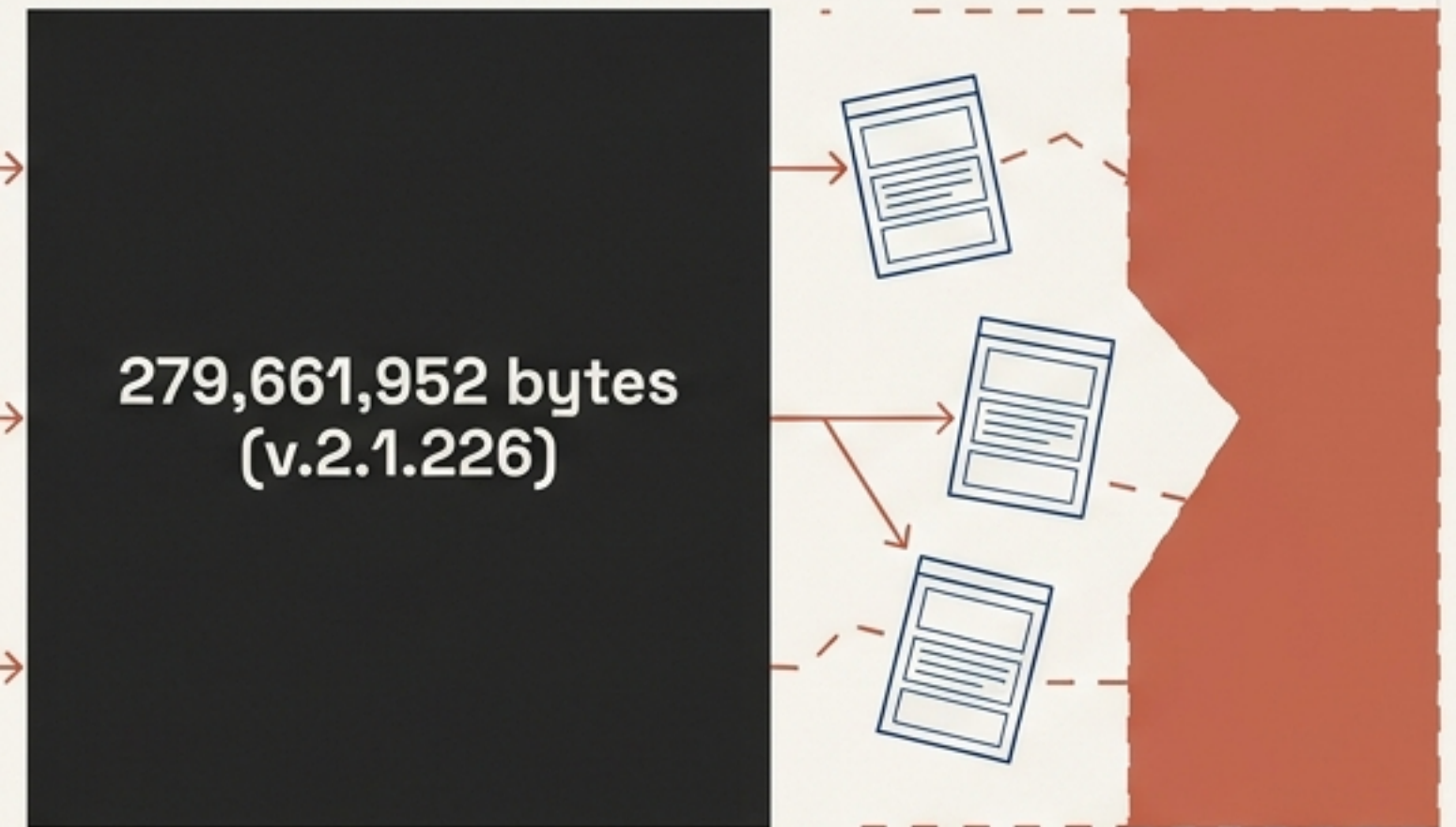
OrchestKit does not have **thirty-four bugs**.
It has **one defect**, duplicated thirty-four times.

The Illusion



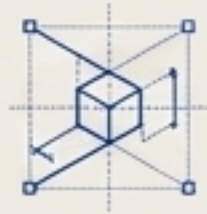
The code is **careful, strictly typed**, and **thoroughly tested**. But it is typed against an imagined API.

The Reality



Across every audited subsystem, **OrchestKit writes to contracts** that nobody **ever verified** against the actual installed Claude Code binary.

Twelve verified instances of the API disconnect.

What OrchestKit Emits 		What Claude Code Actually Reads 
<code>permissionMode</code> in agent frontmatter -----		Ignored entirely for PLUGIN agents
<code>context: inherit</code> on 32 skills -----		Only fork is mapped; inherit is a silent no-op
<code>writes token-budget-state.json</code> -----		The binary writer emits <code>instruction-budget.json</code>
<code>/goal abort-if</code> -----		0 occurrences in the binary, 0 across 357 changelogs

The test suite did not fail to catch the bug; it encoded it.

The screenshot shows a code editor with two panels. The left panel, titled 'Layer 1 | OrchestKit', contains a test suite for 'Permissions Helper'. The right panel, titled 'Layer 2 | Claude Code', shows a memory dump. A green box with a checkmark and '103 passing' is positioned above the test code. A red 'X' is placed over the memory dump, indicating a discrepancy between the test code and the actual data.

Layer 1 | OrchestKit

```
describe("Permissions Helper", () => {
  it("corectly sets permission mode", () => {
    const config = {
      "permissionMode": "755",
    };
    const result = helper.process(config);
    expect(result.permissionMode).toBe("755");
  });
});
```

Layer 2 | Claude Code

Address	Hex	ASCII
0001F4A0	70 65 72 6D 69 73 73 69 6F 5F 6D 6F 64 65	permission_mode.
0001F4B0	37 35 35 00 00 00 00 00 00 00 00 00 00 00	755.....
0001F4C0	55 53 45 5F 43 4F 4E 46 49 47 00 00 00 00	USER_CONFIG....
0001F4D0	01 00 00 00 00 00 00 00 00 00 00 00 00 00	
0001F4A0	70 65 72 6D 69 73 73 69 6F 5F 6D 6F 64 65	permission_mode.
0001F4B0	37 35 35 00 00 00 00 00 00 00 00 00 00 00	755.....
0001F4C0	55 53 45 5F 43 4F 4E 46 49 47 00 00 00 00	USER_CONFIG....
0001F4D0	01 00 00 00 00 00 00 00 00 00 00 00 00 00	

103 unit tests pass on a permission-mode helper that is entirely dead in production. The fixtures were hand-written in camelCase to match the imagined API. Checked against 366,187 strings dumped from the binary, zero camelCase occurrences exist for this key.

An audit that only reports defects cannot be trusted on severity. The foundation is exceptionally strong.

Architecture

152 well-factored hook entries across 29 events. A programmatic diff found **ZERO** dead registrations.

Security Discipline

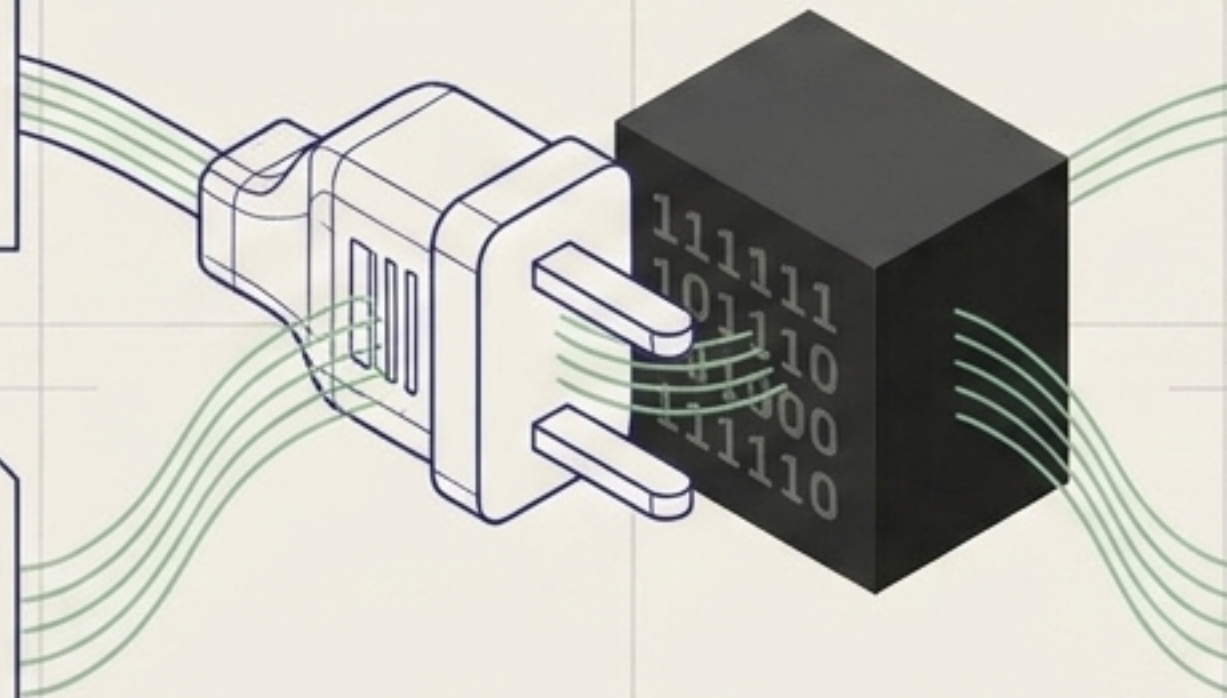
Honest bounds.
network-egress-guard documents itself as bypassable prevention, not containment.

Live Defense

pretool-lesson-guard successfully **caught** the auditor mid-run, **blocking** an evasion command.

Verification Culture

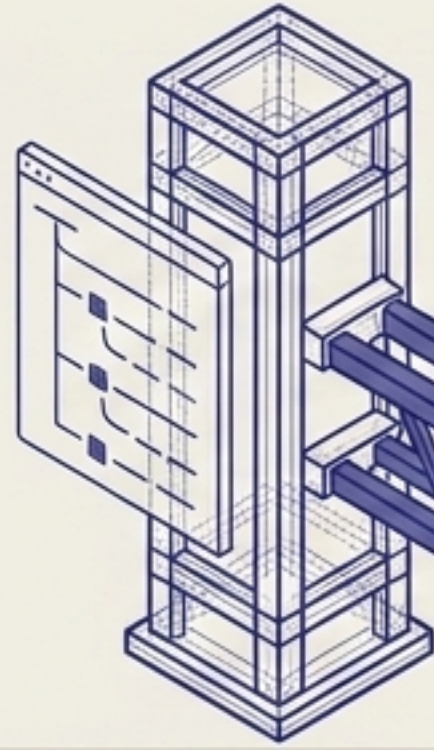
Adversarial-refutation
Adversarial-refutation rules **corrected** the audit itself four times mid-run.



The Real Moat: What the substrate has no mechanism for.

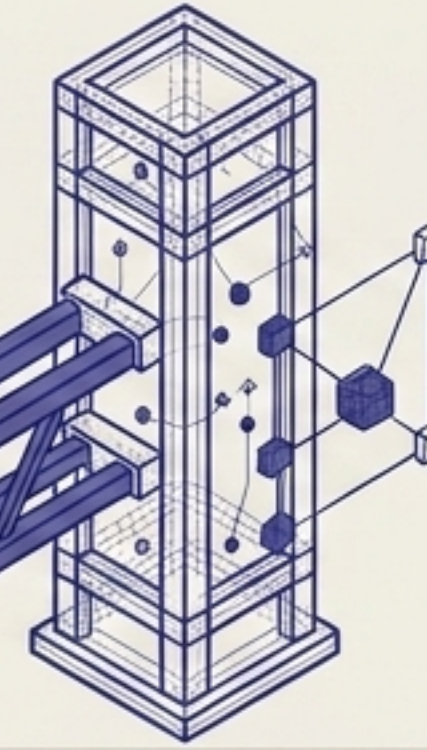
Cross-REPO Orchestration

Opening twelve PRs
across twelve separate
repositories and waiting
on CI.



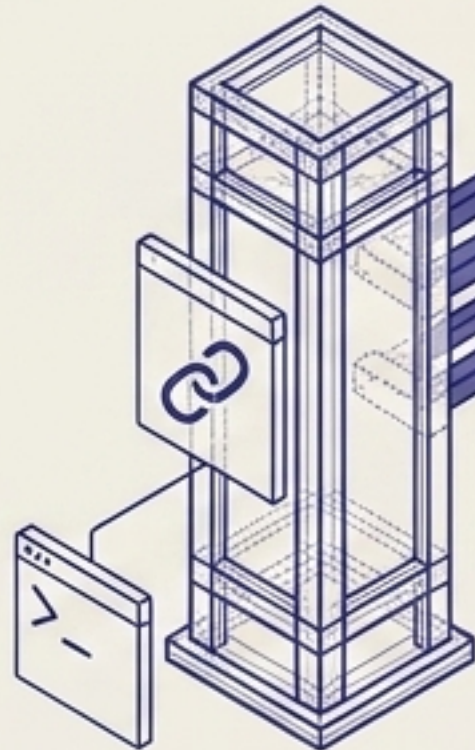
Knowledge-Graph Memory

MCP implementation
with typed entities and
semantic relations.



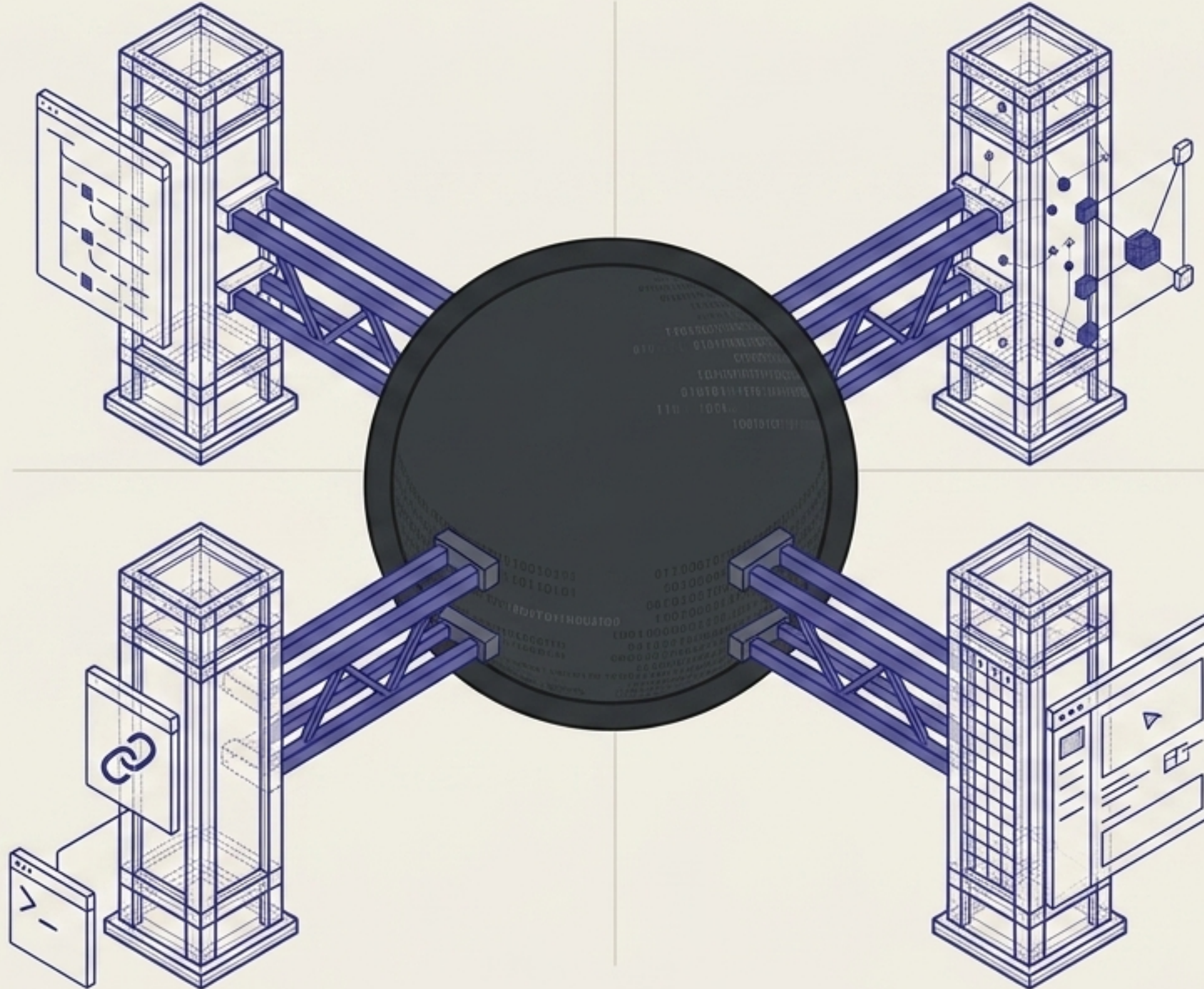
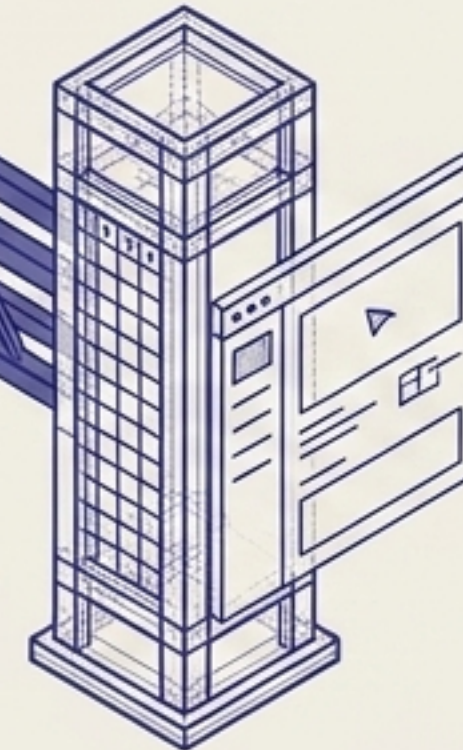
Cross-Model Refuter

Shelling to a non-Claude
CLI to fulfill one quorum
slot.



Visual Standards

The playground standard
and the 618-line
feature-to-version matrix.



Four critical blind spots requiring immediate remediation.

Component	The Illusion 	The Reality 	Impact
State Cleanup	Deleting stale task dirs based on <code>mtime</code> .	Nested file writes don't update parent <code>mtime</code> .	Live teams deleted mid-flight (Fixed in #3305).
Sandbox Policy	Comprehensive allow/deny lists.	The 'enabled' key is completely absent.	3 verified bypasses; nothing blocks SSH key reads.
Goal Syntax	<code>/goal abort-if</code> acts as a safety rail.	Parses as a fresh condition, replacing the first.	Actual acceptance criteria permanently discarded.
Spawn Cap	Context-gate denies overages.	Emits structurally dead <code>PreToolUse</code> on <code>SubagentStart</code> .	Cap has never fired; escape valve checks wrong path.

Context budgets are being paid for a catalog the model cannot read.

499,359

Duplicated characters sitting in 35 generated slash-command wrappers.

29 of 105

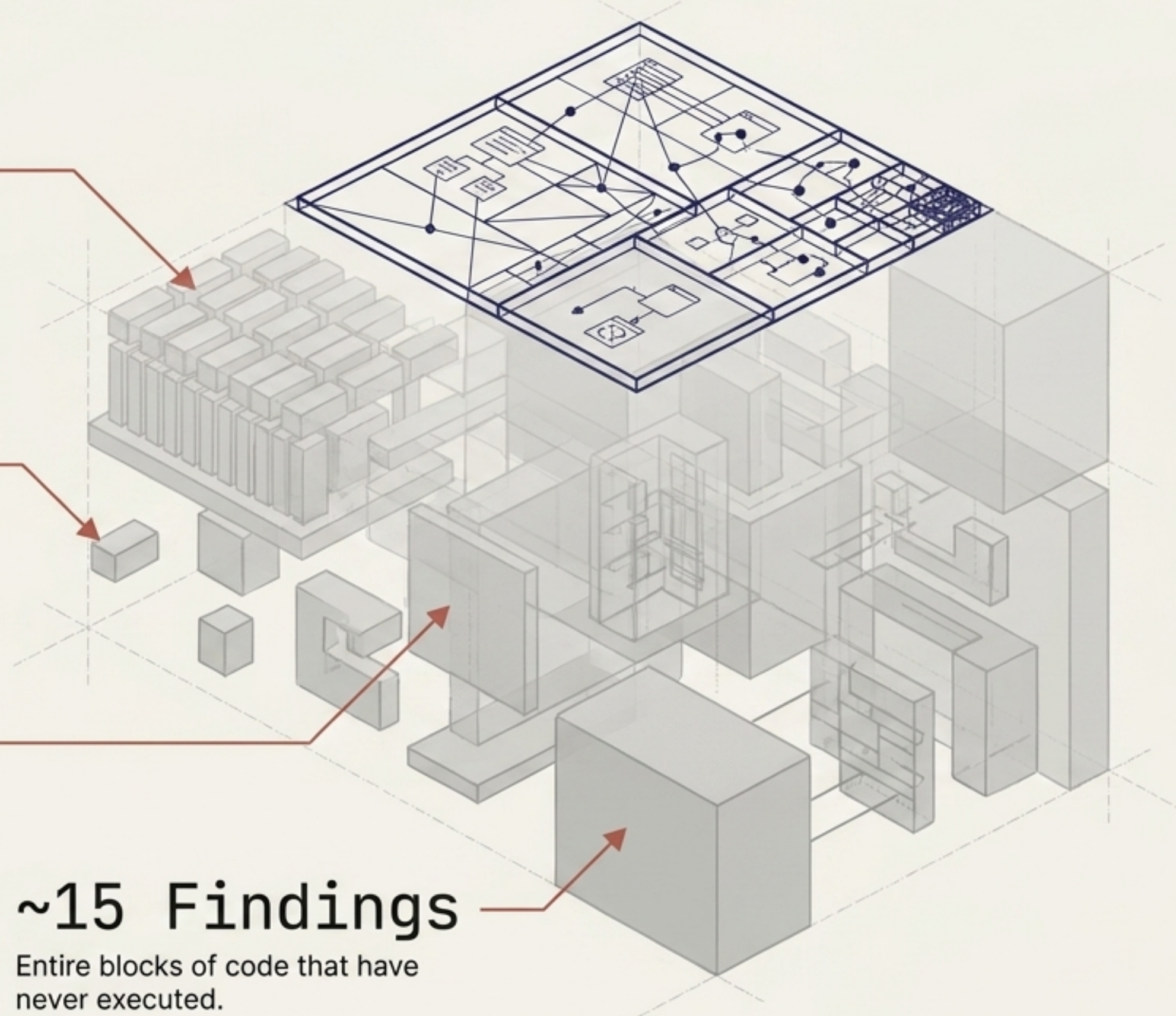
Skills entirely unreachable by every activation path (model-blocked, no preload edges).

369 of 375

Task directories unreachable through the team lookup.

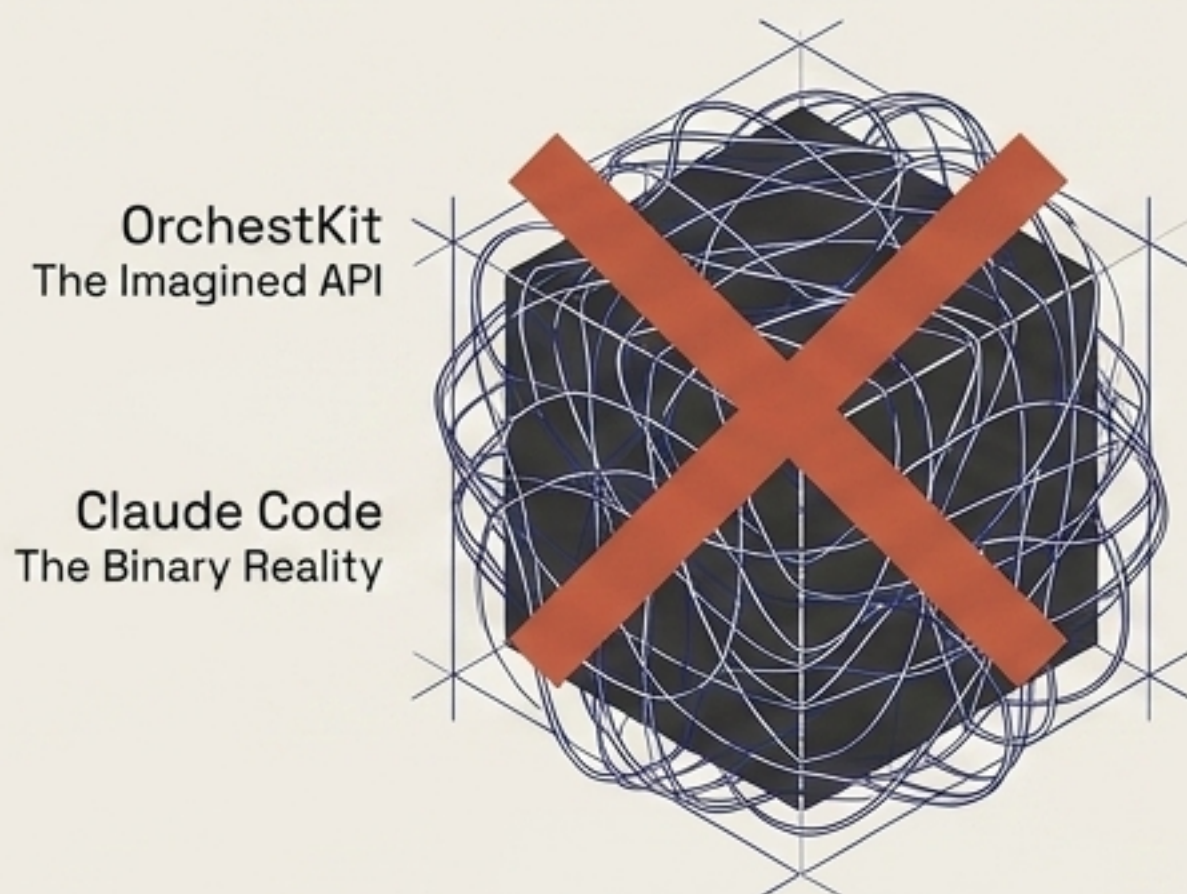
~15 Findings

Entire blocks of code that have never executed.



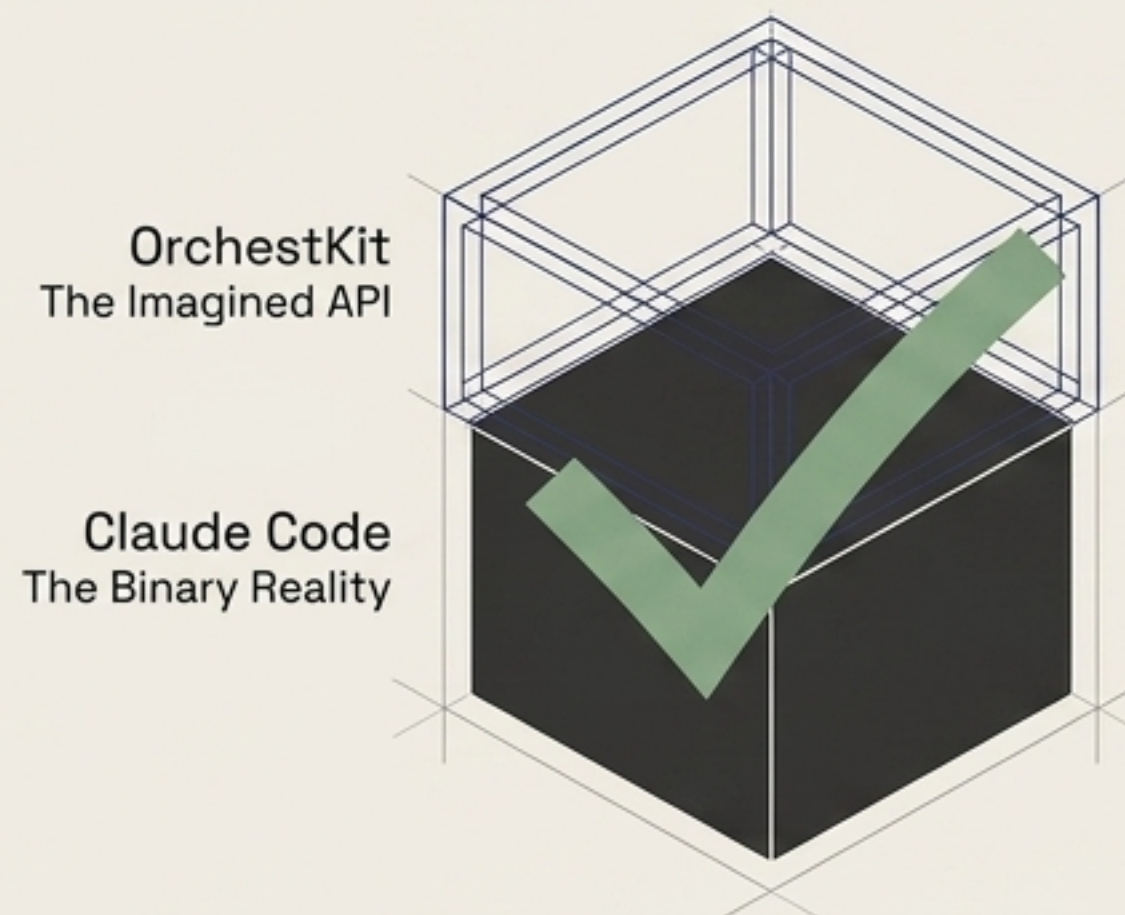
The Governing Rule Violated: A layer on it, never a replacement beside it.

Interception (Violation)



Violation: The binary is enveloped and hidden.

Extension (Target)



Target: The binary is extended and supported.

cmux shim: A 1500-byte bash wrapper intercepting the entrypoint (corrupted three audit buckets by masking the real 279MB binary).

Intercepts and obscures the native entrypoint, hiding the actual binary size and impacting audit accuracy.

Code: ``#!/bin/bash...`` (abbreviated).

worktree provisioner: Silently removes Claude Code's native `.worktreeinclude` baseRef resolution.

Destructively removes critical base reference resolution logic from the native configuration, breaking intended behavior.

Code path: ``.worktreeinclude`` config.

context-gate: Hard-denies where the native runtime caps at 16 concurrent / 1000 per run.

Imposes a rigid denial policy that conflicts with the native runtime's defined concurrency caps.

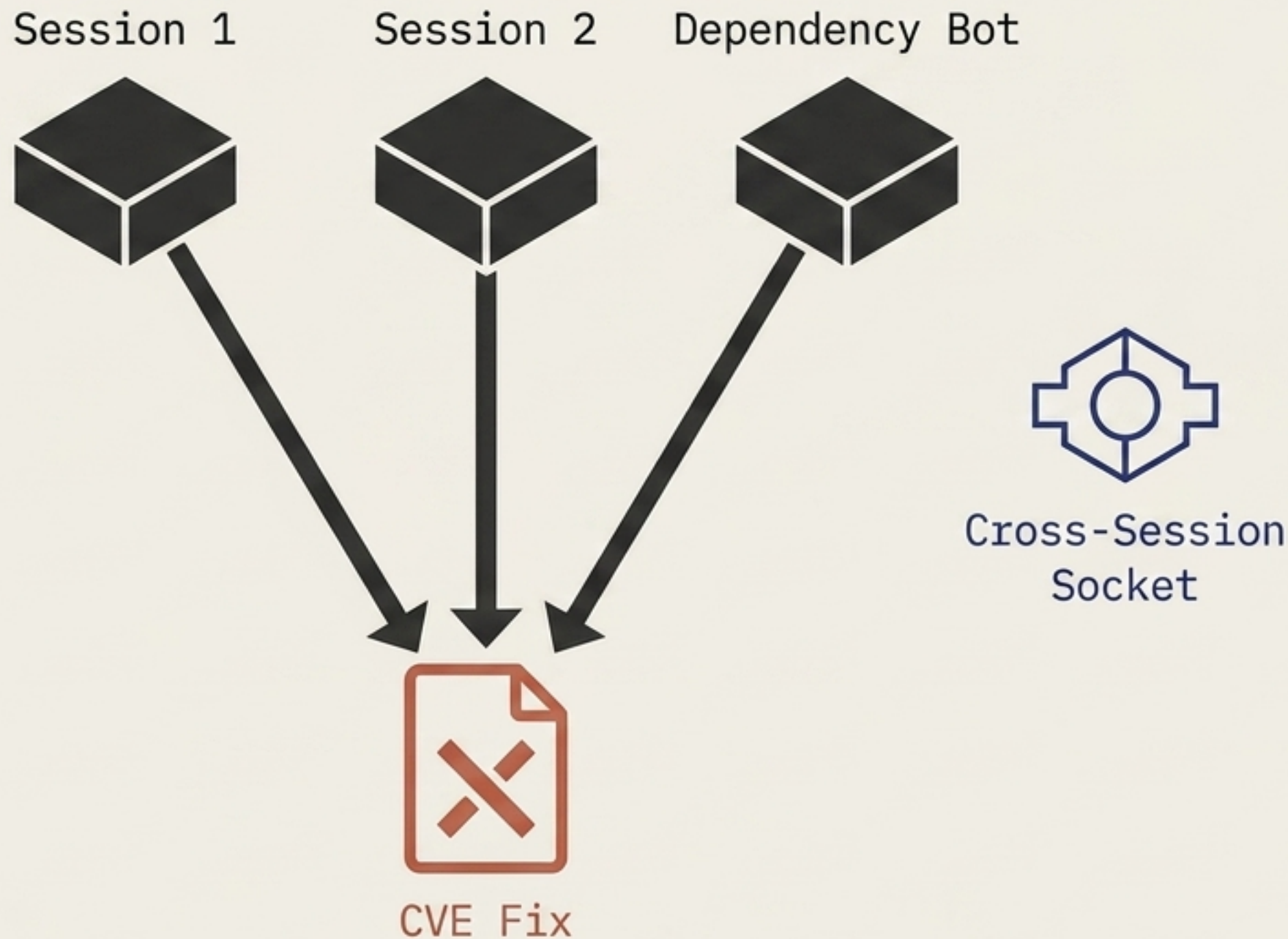
Metrics: ``16 concurrent``, ``1000 per run``.

session-cleanup: Reads and rewrites an internal file explicitly documented as version-unstable.

Interacts with and modifies an internal state file, which is specifically designated as unstable and subject to change.

File: ``INTERNAL_STATE_FILE`` (example).

The thesis proving itself live: The Coordination Failure



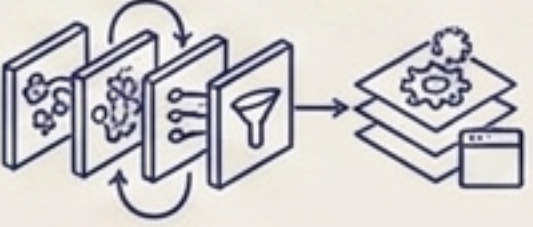
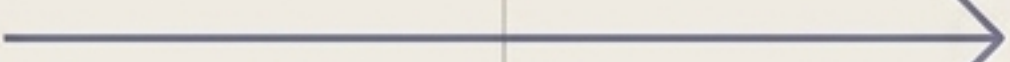
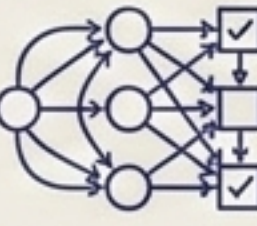
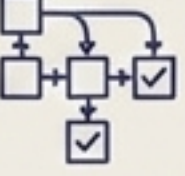
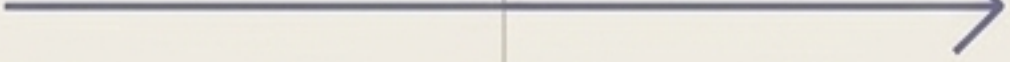
Cross-session messaging is active. The socket is bound.

0 of 152 hooks touch the socket that Claude Code exports before any hook runs.

Result: Six active sessions are mutually unreachable.

Impact: Three independent agents fixed the exact same CVE dependency advisory simultaneously. Two fixes were completely discarded as duplicates.

The pattern is not missing features; it is maintaining machinery the substrate already owns.

OrchestKit Hand-rolled		Retirement / Adoption		Claude Code Native	
Hand-rolled regex guard					Adopt <code>sandbox.enabled</code>
Hand-rolled provisioner					Adopt <code>.worktreeinclude</code>
Prose subagent budget					Adopt Spawn-depth env var
4-hook compaction layer					Adopt Auto-compact window
Hand-rolled dependency checks					Adopt Plugin dependencies array

14 of 39 unused capabilities are high value.

A report that hides its own process errors is not evidence.

~~ci-sentinel uses bare mode.~~

(**Refuted:** dropped flag on 2026-05-18; binds a socket today).

~~The roster exposes session working directories.~~

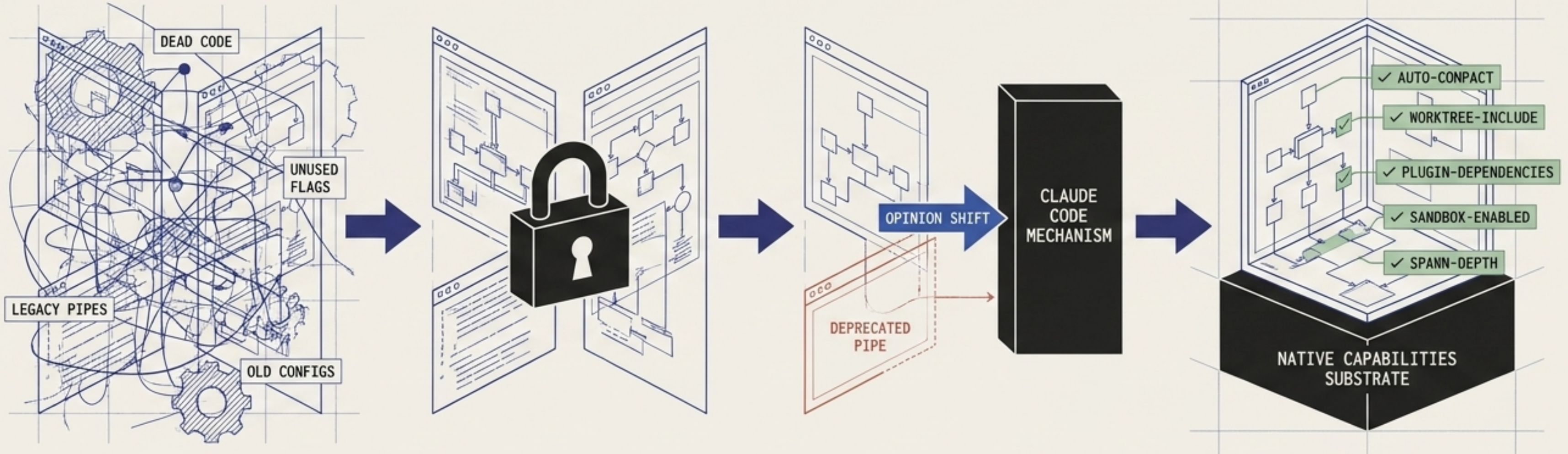
(**Refuted:** live payload returns only name, id, kind, status, age).

~~The branch topology is functional.~~

(**Refuted:** PRs receive 0 of 18 required status checks and sit blocked).

Crucial Blindspot: Token costs were repeatedly asserted as retirement arguments, but never actually measured against the plugin details command.

The remedy is not thirty-four tickets. It is four structural moves.



1. Delete

Shed the dead weight. Remove the ~15 findings of code that has never executed.

2. One Gate

Lock the interface via a single binary contract test (details on next slide).

3. Retire (In Order)

Shift the load. Re-home the opinion onto Claude Code's mechanism before deleting the pipe, preventing lost functionality.

4. Adopt Five

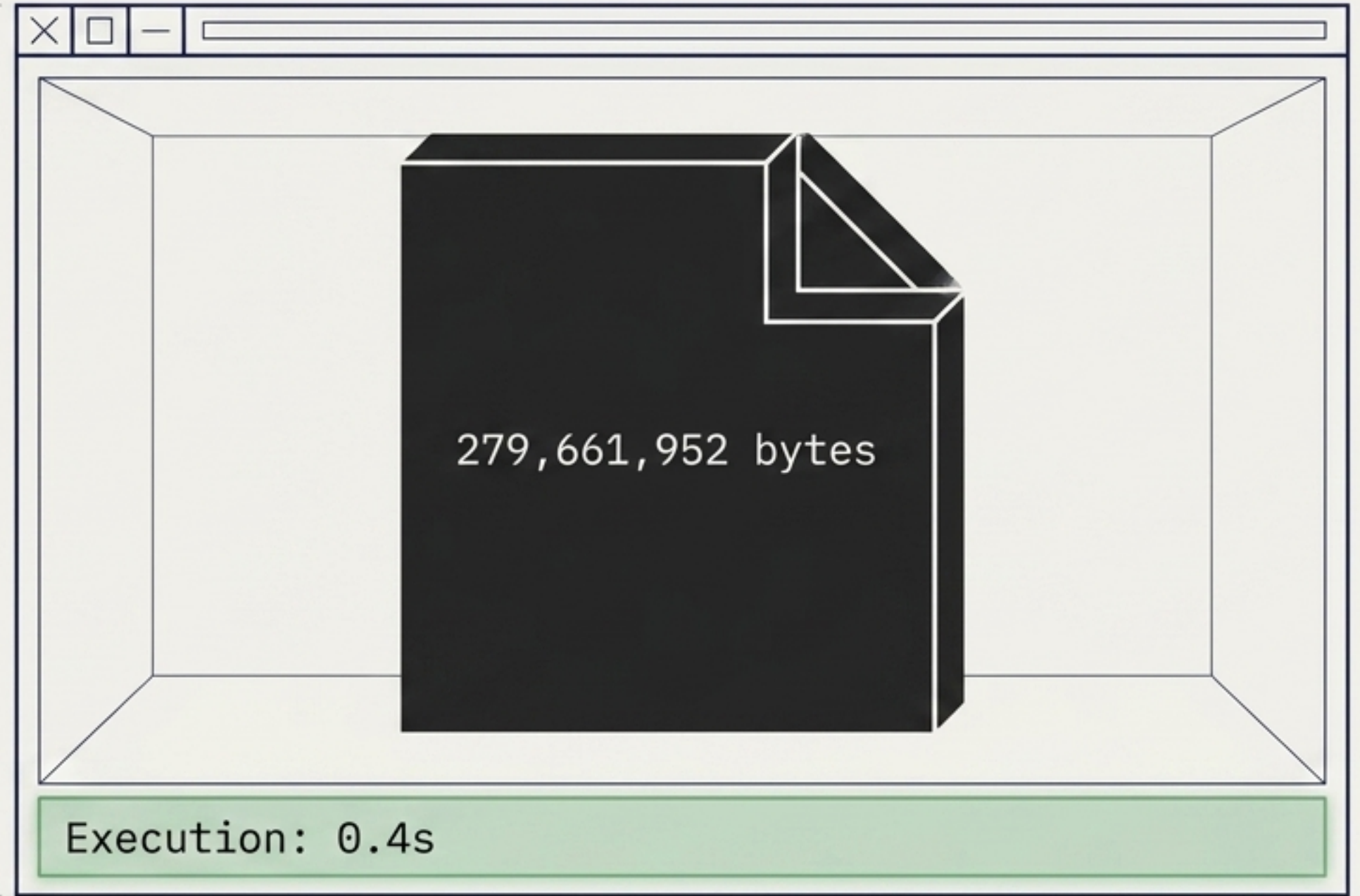
Leverage the substrate. Adopt the 5 native capabilities that delete more maintenance burden than they add.

Move 2: The Single Gate that prevents recurrence.

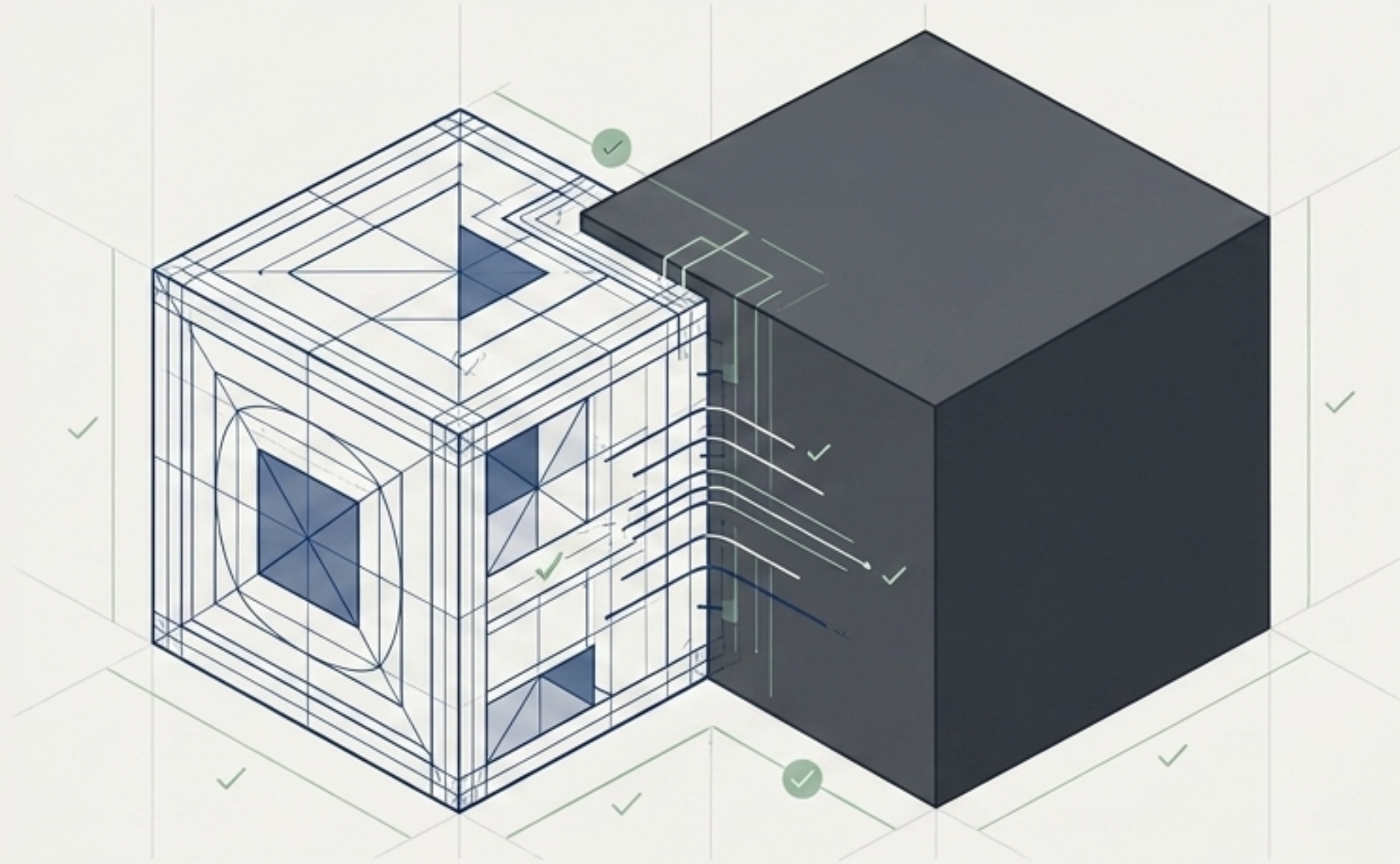
A single contract test must assert that every key OrchestKit emits is a key the binary actually reads.

It operates by grepping the local 279MB binary file directly, running in under a second.

Without this structural gate, fixing the current bugs simply guarantees the next version grows thirty-four more.



True extension. Zero interception. Verified reality.



By adopting the substrate and abandoning the imagined API, OrchestKit sheds its maintenance burden.

This preserves the true Moat—cross-repo orchestration and deep knowledge memory—built firmly on the reality of the binary.